

PAVITHRA BHAT

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PROFESSIONAL SUMMARY

Software Engineer with MERN full-stack experience, building production web, mobile, and AI voice agent features using React.js, Node.js, Express.js, and MongoDB. I am skilled in real-time systems (WebSockets, LiveKit), RAG and LLM integration, REST APIs, and JWT authentication.

SKILLS

Languages: JavaScript, TypeScript, Python, SQL

Frontend: React.js, Next.js, React Native, Redux, HTML5, CSS3, Tailwind CSS

Backend & Databases: Node.js, FastAPI, MongoDB, Supabase, Firebase, REST APIs, WebSockets, JWT

AI & Voice: RAG, LLM API Integration, Prompt Engineering, Embeddings, Vector Databases, Reranking, LlamaIndex, LangChain, Groq, Gemini API, Claude API, STT/TTS, LiveKit, Plivo

Tools: Git, GitHub, Postman, Jira, ClickUp, Vercel, VS Code

PROFESSIONAL EXPERIENCE

Software Engineer | Trikon.Tech, Mangaluru June 2026 – Present

- Built Trikon's AI voice agent on a Node.js/Express backend, integrating STT, LLM and TTS pipelines with Plivo telephony for real-time conversational calls.
- Cut real-time voice latency from 800 ms to 300 ms (62.5%) by redesigning interrupt handling on LiveKit: speech-frame confirmation, turn IDs and silence delay.
- Designed and shipped a React Native mobile app on a Node.js/Express REST API (tested with Postman); delivered production APK and IPA builds.
- Implemented automatic call hangup detection in the LiveKit pipeline; added Supabase-based marketing attribution capture to user onboarding.

Software Development Intern | Design Webtech Pvt Ltd, Bengaluru June 2025 – Nov 2025

- Built and deployed 2 production Android apps using Kotlin, MVVM, Clean Architecture, Retrofit, Room, Coroutines, LiveData and Navigation Components; managed Play Store releases and version control.
- Implemented offline-first functionality, API error handling and Material Design UI; fixed pagination issues in version-compatible API responses.

PROJECTS

Vyce – Enterprise AI Voice Agent Platform | React.js, TypeScript, FastAPI, WebSockets, Vite [Vercel](#)

- Reduced voice-response latency by streaming direct PCM audio buffers over WebSockets instead of standard HTTP audio streaming.
- Built a real-time campaign engine to automate bulk outbound AI calls.
- Added global Pause/Resume/Stop campaign controls using React Portals, stable under high-frequency state changes.

Multi-Source RAG Chat | Python, LlamaIndex, Qdrant, Groq / Claude API, Gradio [GitHub](#)

- Built a RAG chat app that answers questions over PDFs, web pages, YouTube transcripts and text files, with citations (file, page, URL).
- Implemented local embeddings (BAAI/bge-small), Qdrant metadata source filters and follow-up question rewriting using chat history.
- Added optional cross-encoder reranking and an eval script scoring 10 test questions with reranking on and off.

Smart Garbage Collection System | React.js, Node.js, Leaflet.js, Supabase [Vercel](#) | [GitHub](#)

- Built a full-stack app for real-time garbage collection route tracking, visualized on Leaflet.js maps.
- Created a household plastic pickup request module with a user rewards system.
- Developed a RESTful Node.js backend with Supabase for location data and collection scheduling.

EDUCATION

Master of Computer Applications (MCA) | Canara College, Mangaluru | CGPA: 9.0 2024 – 2025

CERTIFICATIONS

- Claude courses (incl. Claude Code in Action) – Anthropic, Skilljar
- Retrieval-Augmented Generation (RAG) Course – Great Learning
- Python for Data Science – NPTEL, IIT Madras (2024)
- React.js – Simplilearn; Web Application Development in React.js – Codelab Systems